MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

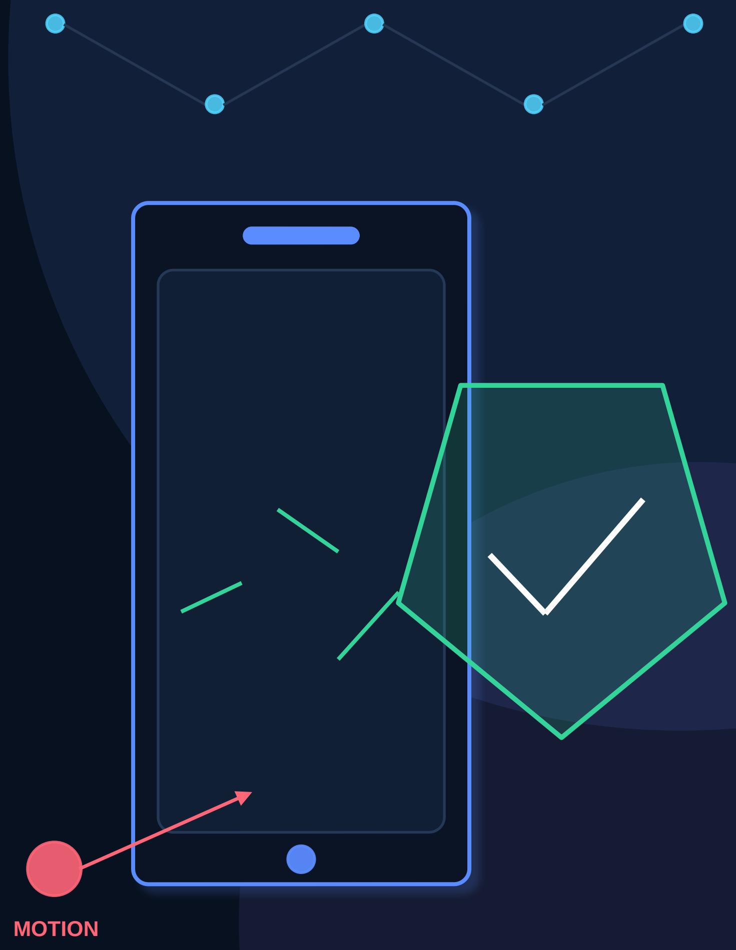
Department of Computer Science and Engineering

PROJECT PRESENTATION

FailureAntyTheft

A phone-based IoT movement detection and anti-theft alert system

Local-first • Real-time • No dedicated sensor hardware



PRESENTED BY

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Why build this system?

Portable assets need monitoring, but dedicated IoT hardware adds cost and setup effort.

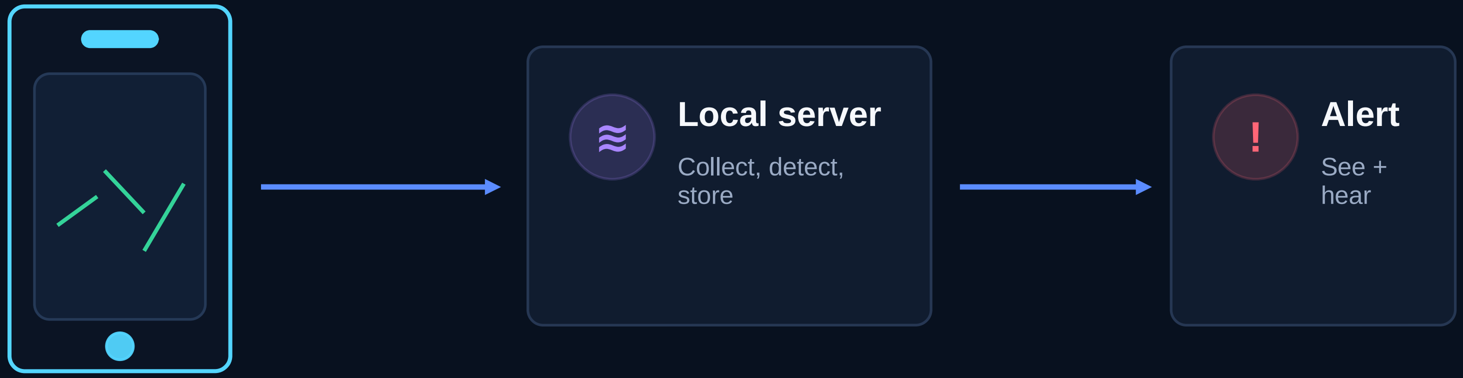
The core problem

- **Portable assets are easy to move**
  Bags, drawers, bicycles and lab equipment may be unattended.
- **Raw sensor noise causes false alarms**
  A useful detector must distinguish noise from sustained movement.
- **Commercial hardware is unnecessary**
  Most smartphones already contain accelerometers, Wi-Fi and batteries.
- **Cloud dependence adds friction**
  A classroom demo should work privately on a local network.

Opportunity

Reuse hardware people already own

Turn a phone into a sensor node and a laptop into the complete local monitoring platform.



LOW COST | PRIVATE LAN | REUSABLE

One phone. One local server. One clear response.

FailureAntyTheft covers the complete IoT loop: sense, communicate, process, store, visualize and act.

≈ 1 SENSE

Phyphox exposes live X, Y and Z acceleration plus device time from each phone.

◇ 2 DECIDE

A deterministic detector calibrates a baseline, filters noise and verifies sustained movement.

! 3 RESPOND

The dashboard updates in real time, persists the event and plays a repeating siren.

DESIGN PRINCIPLE

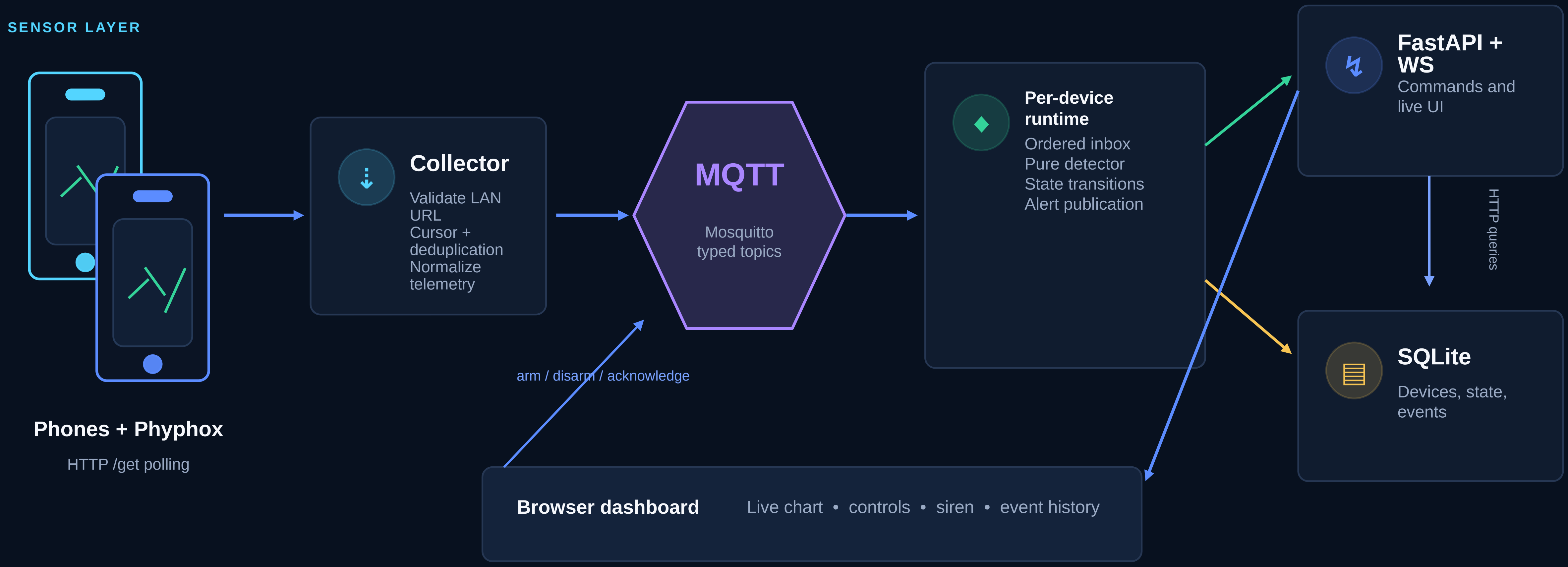
Local-first by default

No paid cloud, no custom mobile app and no silent fallback when MQTT is configured.

MULTI-PHONE · REAL-TIME · AUDITABLE

A modular local IoT pipeline

Typed messages separate collection, detection, persistence and presentation while one FastAPI process supervises the application.



What happens after the phone moves?

Every stage removes ambiguity before the operator is interrupted.

01 → 02 → 03 → 04 → 05 → 06

01 Sample
Phyphox emits time + X/Y/Z

02 Collect
Poll, validate and deduplicate

03 Normalize
Publish typed telemetry

04 Score
Magnitude → EMA → baseline delta

05 Confirm
3 crossings inside 0.5 s

06 Alert
Persist, push UI and sound siren

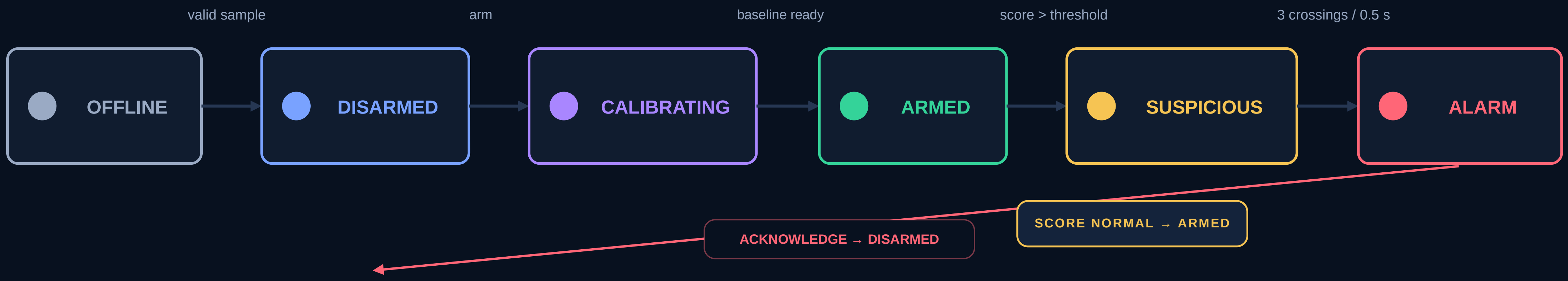
WHY DEVICE TIME MATTERS

Buffered samples are judged by the phone’s sample time—not by clustered HTTP arrival time.

This preserves the sustained-movement rule and prevents network timing from changing the detector’s decision.

Calibrate first. Filter noise. Confirm sustained movement.

The detector is a pure, replayable state machine with no network, database or clock dependencies.



CALIBRATION

5 s placement delay + 2 s sampling window

Median magnitude becomes the baseline; excessive variance rejects calibration.

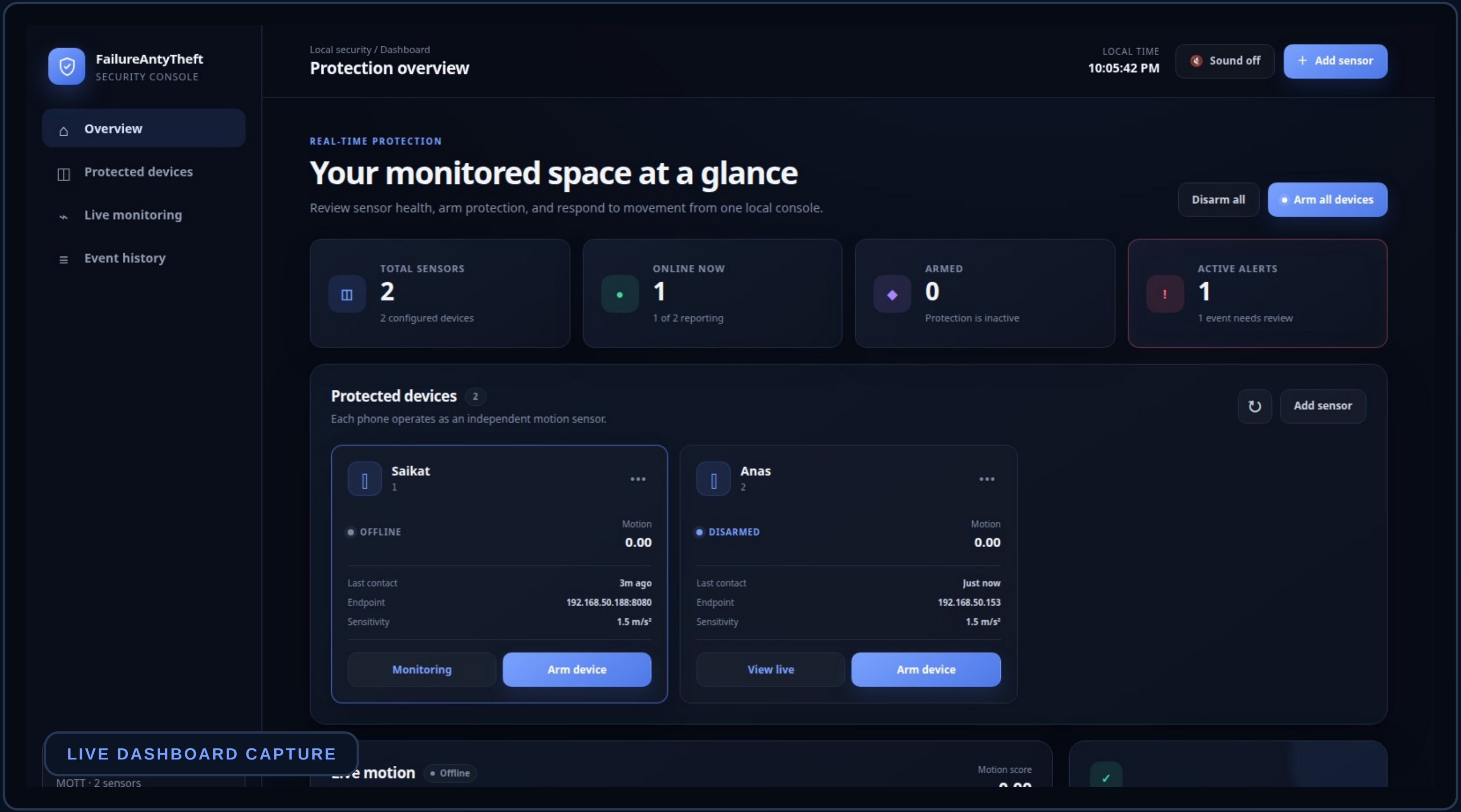
MOTION SCORE

$$\text{score} = \left| \mathrm{EMA}\left(\sqrt{x^2 + y^2 + z^2}\right) - \text{baseline} \right|$$

Sensitivity sets the threshold; $\alpha = 0.35$ smooths short sensor noise.

A professional real-time security console

The interface turns low-level sensor data into one clear operational view.



LIVE DASHBOARD CAPTURE

Designed for fast decisions

- **At-a-glance health**
  Online, armed and active-alert counts.
- **Independent sensors**
  Arm, disarm, edit and monitor each phone.
- **Live WebSocket chart**
  Selected-device motion score without raw-data overload.
- **Actionable alarm**
  Visual banner, repeating siren and acknowledge workflow.
- **Auditable history**
  Persistent events with CSV export.

Simple tools, clear responsibilities

Each technology was selected for local reliability, testability and low operational cost.

P
PHONE SENSOR
Phyphox
Accelerometer access + remote HTTP API

Py
BACKEND
Python + FastAPI + asyncio
Lifecycle, REST commands and WebSocket delivery

M
MESSAGING
MQTT + Mosquitto
Typed publish/subscribe topics and reconnect behavior

DB
PERSISTENCE
SQLite (WAL)
Device configuration, runtime state and event history

UI
FRONT END
HTML + CSS + JavaScript
Responsive dashboard, canvas chart and Web Audio siren

✓
QUALITY
Pytest + Ruff + mypy + Docker
Replay tests, static checks and reproducible broker

Deployment: one laptop runs the application, Mosquitto, SQLite and dashboard; phones remain sensor nodes on the same private LAN.

Verified in tests—and against a real phone

Automated replay protects detector behavior while live checks prove the Phyphox integration path.

105
tests passed
1 environment-gated skip

84%
coverage at handoff
Core detector: 94%

200
live /config status
Phone measuring = true

4
verified buffers
acc_time + accX/Y/Z

Test strategy

- **Pure detector unit tests**
  Calibration, noise filtering, crossings and transitions.
- **Recorded-shape replay**
  Stationary data stays quiet; movement triggers exactly once.
- **Integration checks**
  Repository, API, runtime, WebSocket assets and live MQTT round-trip.

LIVE PHONE RESULT

Saikat DISARMED • ONLINE

The collector’s default time buffer was corrected from t to acc_time using the phone’s live /config evidence.

A seven-step proof in under two minutes

Use a private hotspot or Wi-Fi network where the phone and laptop can communicate directly.

1 **START**

Open Acceleration with g and enable remote access

2 **CONNECT**

Register the displayed private URL

3 **VERIFY**

Device appears Disarmed—not Offline

4 **SOUND**

Click Sound off and hear the test siren

5 **ARM**

Keep still for 5 s placement + 2 s calibration

6 **MOVE**

Move the phone; observe Alarm + siren

7 **CLOSE**

Acknowledge, disarm and show event history

DEMO SAFETY NET Keep a screenshot and event-history explanation ready if venue Wi-Fi isolates devices.

What the prototype does—and what comes next

The current system is presentation-ready on a trusted LAN, but it is not a certified security product.

CURRENT LIMITATIONS

- **Trusted private LAN only**
  No authentication or TLS for public deployment.
- **Browser is the alarm endpoint**
  The tab must remain open with audio enabled.
- **Default Phyphox profile**
  Custom experiments may expose different buffer names.
- **Prototype acceptance scope**
  One live phone verified; multi-phone physical stress testing remains.
- **Not theft prevention**
  It detects and reports movement; it does not physically secure an asset.

ROADMAP

01	**Auto-discover buffers**	Read /config and map experiments automatically.
02	**Secure remote access**	Authentication, HTTPS and role-based controls.
03	**Mobile/PWA notifications**	Background push when the dashboard is not visible.
04	**Multi-sensor correlation**	Combine simultaneous movement to reduce false alarms.
05	**Packaged deployment**	Install as a background service with guided setup.

CONCLUSION

A complete IoT loop—built from devices we already own

FailureAntyTheft proves that a smartphone, a local network and a carefully designed detector can deliver practical real-time monitoring without dedicated hardware or paid cloud services.

105
passing tests

REAL-TIME
WebSocket updates

LOCAL
privacy by default

AUDIBLE
actionable alarm

Questions?

Thank you